【1】R22　冷水机组 / Chiller　Te = 5℃，Tc = 45℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

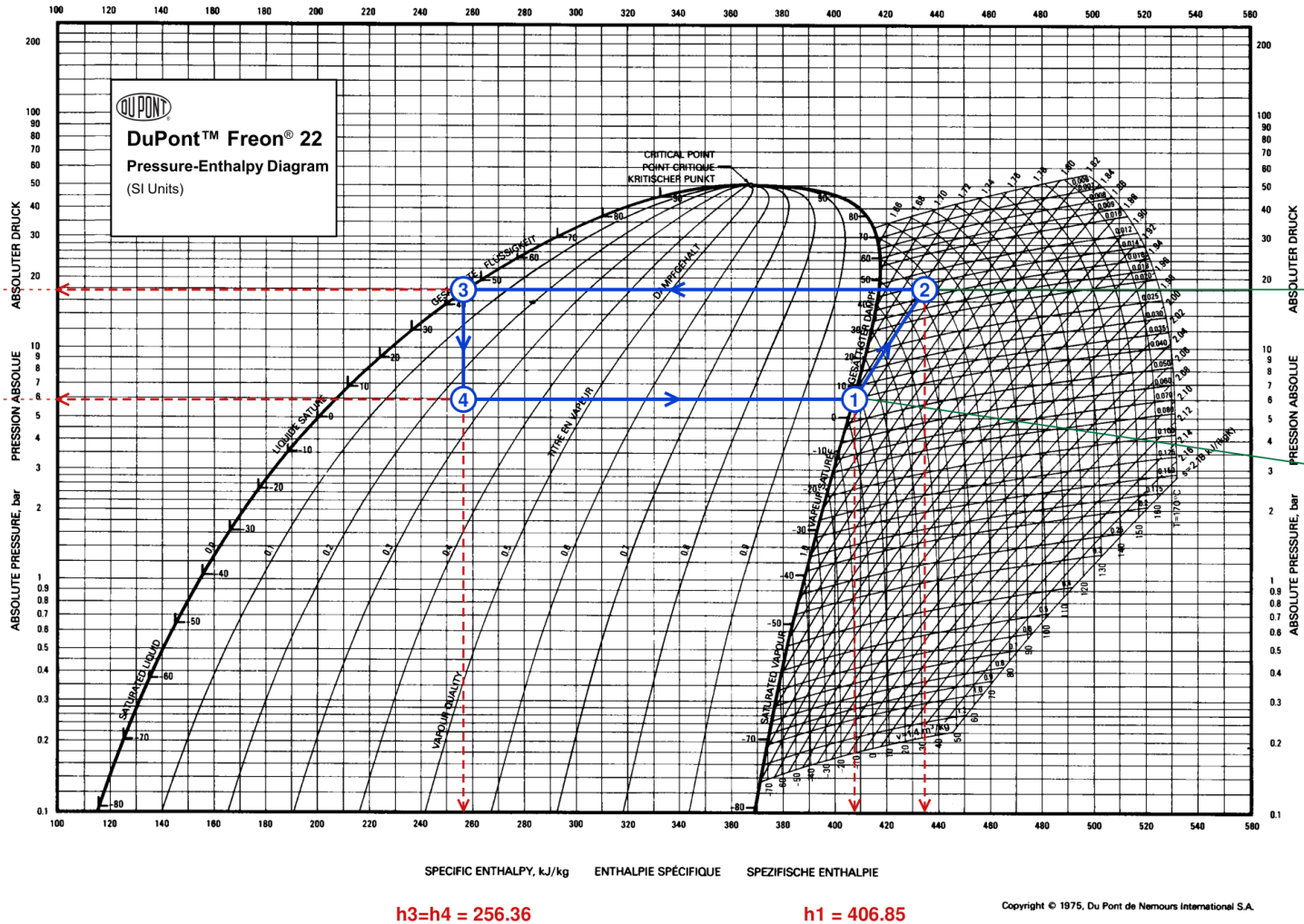
BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.

R22 | Chiller
Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.

pc = 17.292
bar (absolute)

pe = 5.841
bar (absolute)



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POINT 2: COMPRESSOR OUT
v2 = 0.01501 m3/kg
s2 = 1.74361 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.04034 m3/kg
s1 = 1.74361 kJ/(kg.K)

h3=h4 = 256.36

h1 = 406.85

h2 = 433.82

FROM h READINGS
qe = h1-h3 = 150.49
w = h2-h1 = 26.97
COP = qe/w = 5.58
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

【2】R22　房间冷却器 / Room cooler　Te = 10℃，Tc = 60℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

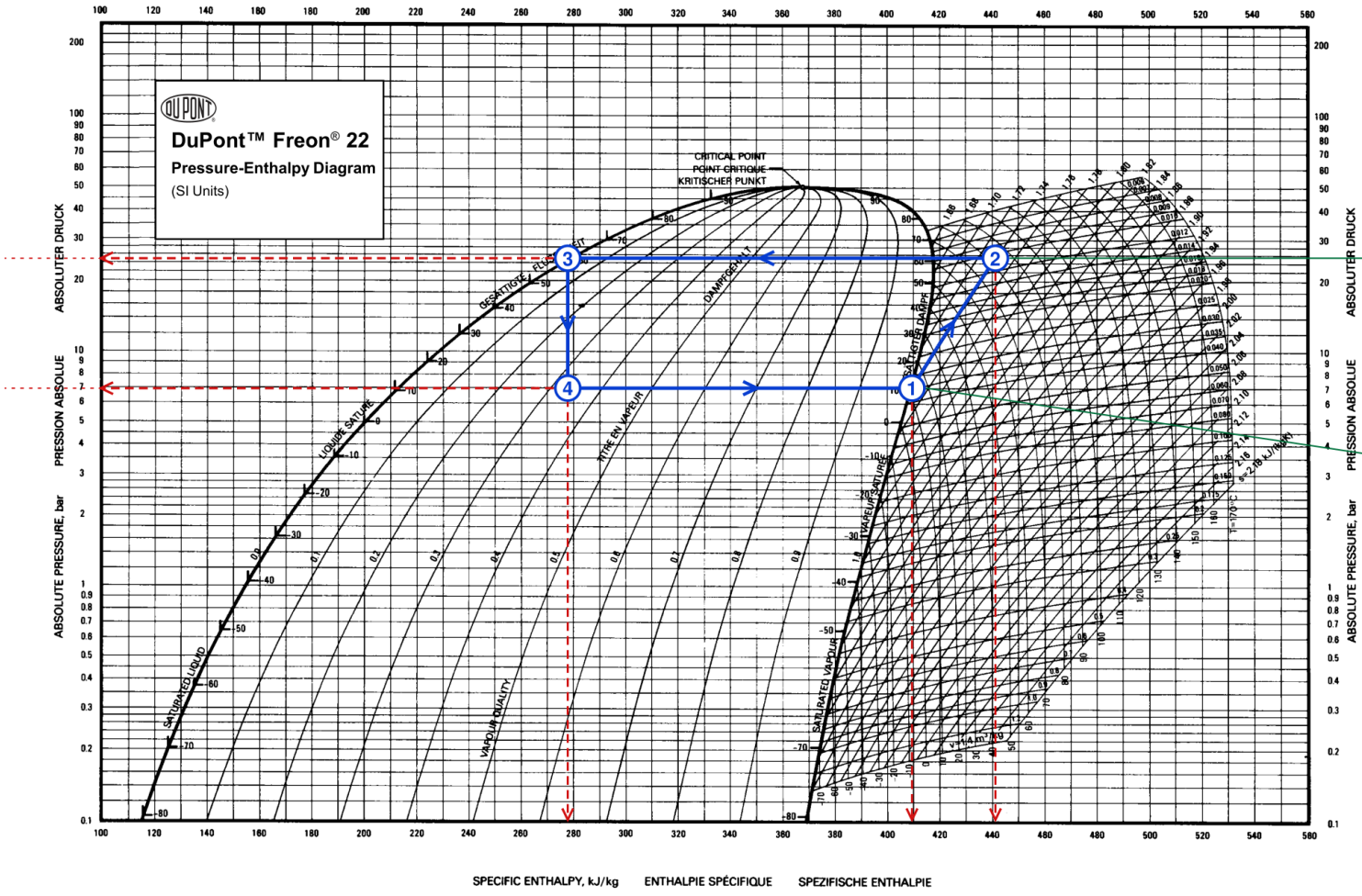
BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.

R22 | Room cooler
Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.

pc = 24.275
bar (absolute)

pe = 6.809
bar (absolute)



POINT 2: COMPRESSOR OUT
v2 = 0.01071 m3/kg
s2 = 1.73680 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.03470 m3/kg
s1 = 1.73680 kJ/(kg.K)

h3=h4 = 277.61

h1 = 408.56

h2 = 440.29

FROM h READINGS
qe = h1-h3 = 130.95
w = h2-h1 = 31.73
COP = qe/w = 4.13
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

R22　读图数据与逐步计算 / Readings and calculations

计算基准：1 为饱和蒸气，3 为饱和液体；1→2 等熵，h4 = h3；无额外过热／过冷。压力均为绝对压力。

Basis: saturated vapour at 1, saturated liquid at 3; 1→2 isentropic; h4 = h3; no extra superheat/subcooling. Pressures are absolute.

读数及结果 / Quantity	**冷水机组 / Chiller** **5℃ / 45℃**	**房间冷却器 / Room cooler** **10℃ / 60℃**
蒸发压力 / Evaporating pressure p_e / bar abs	5.841	6.809
冷凝压力 / Condensing pressure p_c / bar abs	17.292	24.275
入口焓 / Suction enthalpy h1 / kJ·kg^{-1}	406.85	408.56
出口焓 / Discharge enthalpy h2 / kJ·kg^{-1}	433.82	440.29
液体及节流出口焓 / Liquid and throttle outlet h3 = h4 / kJ·kg^{-1}	256.36	277.61
入口比容 / Inlet specific volume v1 / m^3·kg^{-1}	0.04034	0.03470
出口比容 / Outlet specific volume v2 / m^3·kg^{-1}	0.01501	0.01071
入口、出口熵 / Inlet and outlet entropy s1 = s2 / kJ·kg^{-1}·K^{-1}	1.74361	1.73680
制冷效应 / Refrigeration effect q_e / kJ·kg^{-1}	150.49	130.95
单位质量压缩功 / Specific compression work w / kJ·kg^{-1}	26.97	31.73
制冷性能系数 / Coefficient of performance COP	5.58	4.13

代入计算 / Substitution

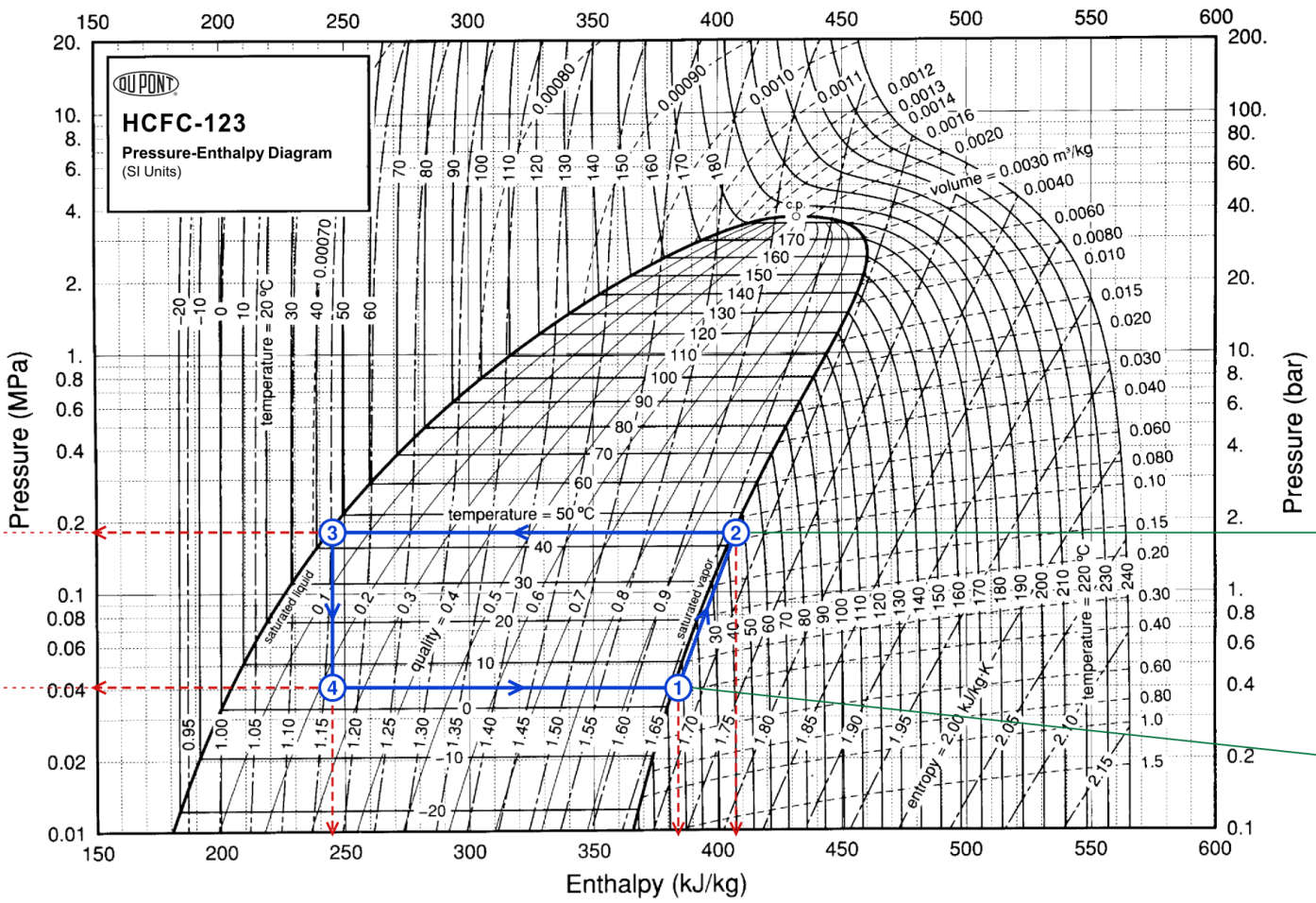
计算 / Calculation	**Chiller**	**Room cooler**
⑤ 制冷效应 / Refrigeration effect q_e = h1 − h3	406.85 - 256.36 = 150.49 kJ/kg	408.56 - 277.61 = 130.95 kJ/kg
⑥ 压缩功 / Compressor work w = h2 − h1	433.82 - 406.85 = 26.97 kJ/kg	440.29 - 408.56 = 31.73 kJ/kg
⑦ COP COP = q_e / w	150.49 / 26.97 = 5.58	130.95 / 31.73 = 4.13

图为近似读数，物性辅助核对；计算用表列焓值。 / Approximate plot readings, property checked; arithmetic uses the displayed enthalpies.

【3】R123　冷水机组 / Chiller　Te = 5℃，Tc = 45℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.



pc = 0.1817
MPa / 1.817 bar

pe = 0.0408
MPa / 0.408 bar

h3=h4 = 245.81

h1 = 384.44

h2 = 407.57

R123 | Chiller
Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.
Ideal point 2 is slightly
inside the saturation dome.

POINT 2: COMPRESSOR OUT
v2 = 0.08853 m3/kg
s2 = 1.66325 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.36203 m3/kg
s1 = 1.66325 kJ/(kg.K)

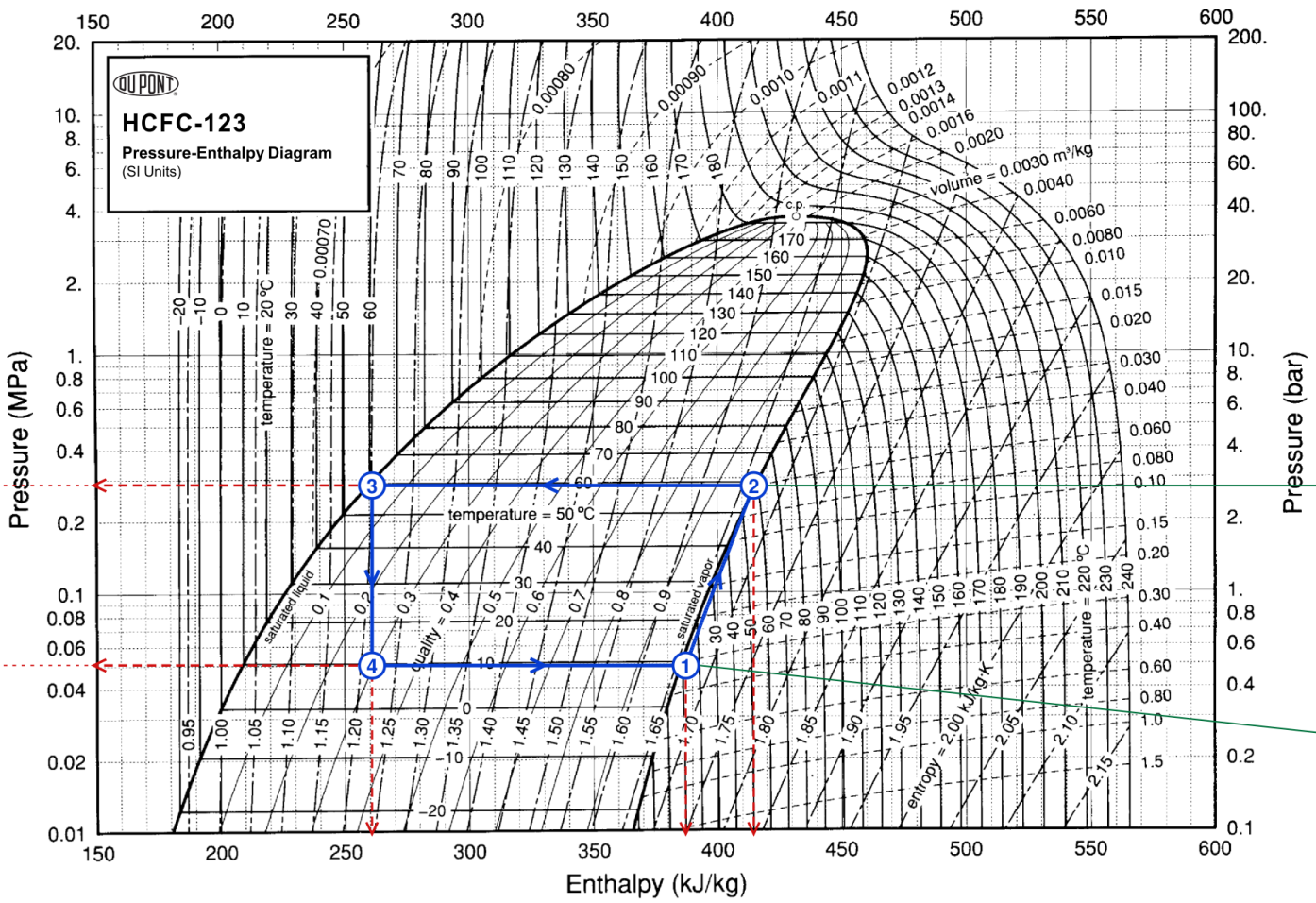
FROM h READINGS
qe = h1-h3 = 138.63
w = h2-h1 = 23.13
COP = qe/w = 5.99
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

【4】R123　房间冷却器 / Room cooler　Te = 10℃，Tc = 60℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.



pc = 0.2859
MPa / 2.859 bar

pe = 0.0506
MPa / 0.506 bar

h3=h4 = 261.67

h1 = 387.46

h2 = 414.71

R123 | Room cooler
Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.
Ideal point 2 is slightly
inside the saturation dome.

POINT 2: COMPRESSOR OUT
v2 = 0.05678 m3/kg
s2 = 1.66264 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.29637 m3/kg
s1 = 1.66264 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 125.79
w = h2-h1 = 27.25
COP = qe/w = 4.62
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

R123　读图数据与逐步计算 / Readings and calculations

计算基准：1 为饱和蒸气，3 为饱和液体；1→2 等熵，h4 = h3；无额外过热／过冷。压力均为绝对压力。

Basis: saturated vapour at 1, saturated liquid at 3; 1→2 isentropic; h4 = h3; no extra superheat/subcooling. Pressures are absolute.

读数及结果 / Quantity	冷水机组 / Chiller 5℃ / 45℃	房间冷却器 / Room cooler 10℃ / 60℃
蒸发压力 / Evaporating pressure p_e / bar abs	0.408	0.506
冷凝压力 / Condensing pressure p_c / bar abs	1.817	2.859
入口焓 / Suction enthalpy h1 / kJ·kg^{-1}	384.44	387.46
出口焓 / Discharge enthalpy h2 / kJ·kg^{-1}	407.57	414.71
液体及节流出口焓 / Liquid and throttle outlet h3 = h4 / kJ·kg^{-1}	245.81	261.67
入口比容 / Inlet specific volume v1 / m^3·kg^{-1}	0.36203	0.29637
出口比容 / Outlet specific volume v2 / m^3·kg^{-1}	0.08853	0.05678
入口、出口熵 / Inlet and outlet entropy s1 = s2 / kJ·kg^{-1}·K^{-1}	1.66325	1.66264
制冷效应 / Refrigeration effect q_e / kJ·kg^{-1}	138.63	125.79
单位质量压缩功 / Specific compression work w / kJ·kg^{-1}	23.13	27.25
制冷性能系数 / Coefficient of performance COP	5.99	4.62

代入计算 / Substitution

计算 / Calculation	Chiller	Room cooler
⑤ 制冷效应 / Refrigeration effect q_e = h1 − h3	384.44 - 245.81 = 138.63 kJ/kg	387.46 - 261.67 = 125.79 kJ/kg
⑥ 压缩功 / Compressor work w = h2 − h1	407.57 - 384.44 = 23.13 kJ/kg	414.71 - 387.46 = 27.25 kJ/kg
⑦ COP COP = q_e / w	138.63 / 23.13 = 5.99	125.79 / 27.25 = 4.62

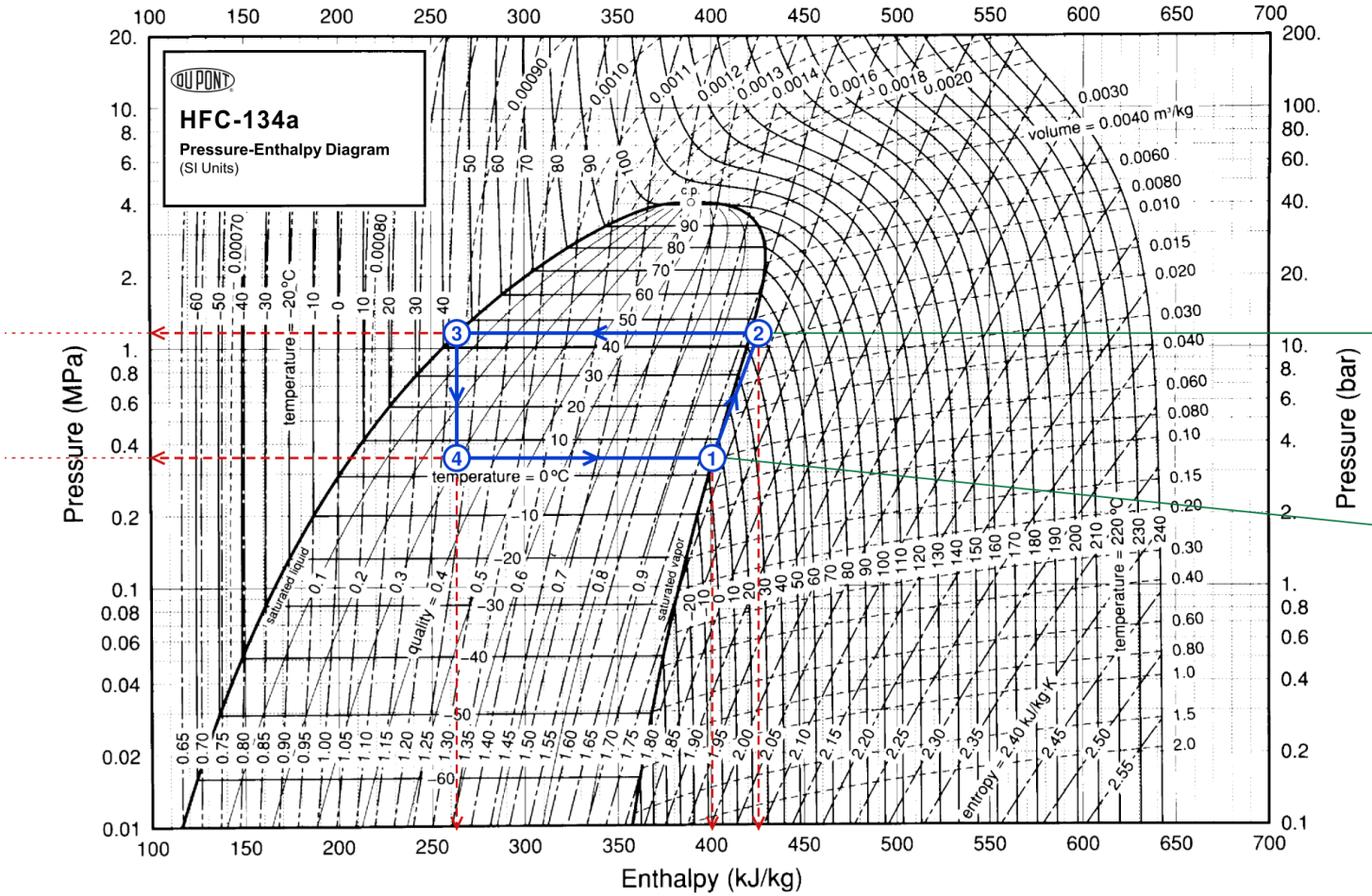
R123：点 2 略入两相区（理想饱和吸气基准）。 / Point 2 is slightly inside the saturation dome on this ideal saturated-suction basis.

图为近似读数，物性辅助核对；计算用表列焓值。 / Approximate plot readings, property checked; arithmetic uses the displayed enthalpies.

【5】R134a　冷水机组 / Chiller　Te = 5℃，Tc = 45℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.



pc = 1.1599
MPa / 11.599 bar

pe = 0.3497
MPa / 3.497 bar

R134a | Chiller
Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.

POINT 2: COMPRESSOR OUT
v2 = 0.01790 m3/kg
s2 = 1.72446 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.05837 m3/kg
s1 = 1.72446 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 137.55
w = h2-h1 = 24.91
COP = qe/w = 5.52
qe and w: kJ/kg

h3=h4 = 263.94　　**h1 = 401.49**

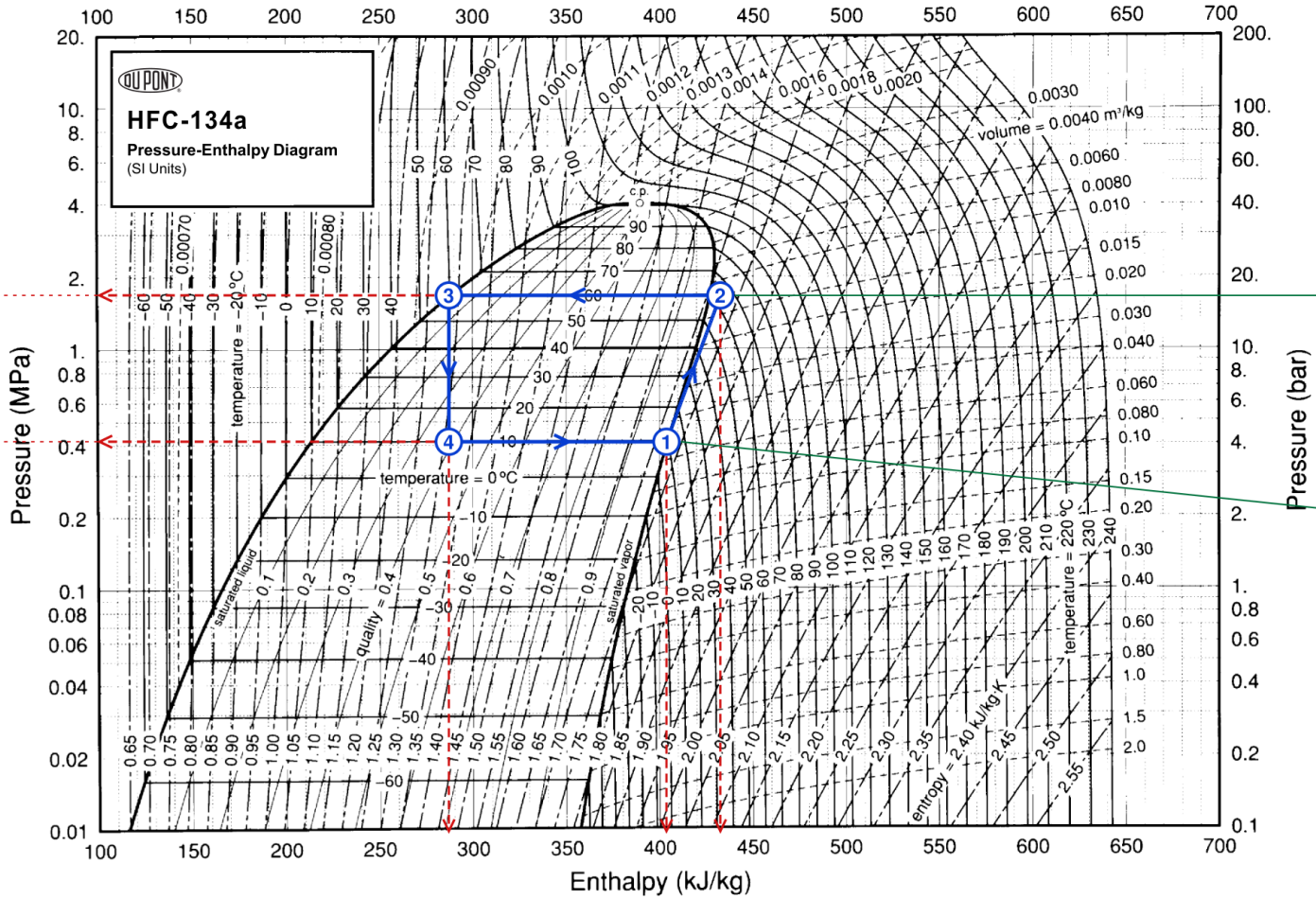
h2 = 426.40

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

【6】R134a　房间冷却器 / Room cooler　Te = 10℃，Tc = 60℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.



pc = 1.6818
MPa / 16.818 bar

pe = 0.4146
MPa / 4.146 bar

h3=h4 = 287.50　　**h1 = 404.32**

h2 = 433.24

R134a | Room cooler
Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2
Read v1 and v2 at points 1 and 2.

POINT 2: COMPRESSOR OUT
v2 = 0.01200 m3/kg
s2 = 1.72211 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.04944 m3/kg
s1 = 1.72211 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 116.82
w = h2-h1 = 28.92
COP = qe/w = 4.04
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

R134a 读图数据与逐步计算 / Readings and calculations

计算基准：1 为饱和蒸气，3 为饱和液体；1→2 等熵，h4 = h3；无额外过热／过冷。压力均为绝对压力。

Basis: saturated vapour at 1, saturated liquid at 3; 1→2 isentropic; h4 = h3; no extra superheat/subcooling. Pressures are absolute.

读数及结果 / Quantity	**冷水机组 / Chiller** **5℃ / 45℃**	**房间冷却器 / Room cooler** **10℃ / 60℃**
蒸发压力 / Evaporating pressure p_e / bar abs	3.497	4.146
冷凝压力 / Condensing pressure p_c / bar abs	11.599	16.818
入口焓 / Suction enthalpy h1 / kJ·kg^{-1}	401.49	404.32
出口焓 / Discharge enthalpy h2 / kJ·kg^{-1}	426.40	433.24
液体及节流出口焓 / Liquid and throttle outlet h3 = h4 / kJ·kg^{-1}	263.94	287.50
入口比容 / Inlet specific volume v1 / m^3·kg^{-1}	0.05837	0.04944
出口比容 / Outlet specific volume v2 / m^3·kg^{-1}	0.01790	0.01200
入口、出口熵 / Inlet and outlet entropy s1 = s2 / kJ·kg^{-1}·K^{-1}	1.72446	1.72211
制冷效应 / Refrigeration effect q_e / kJ·kg^{-1}	137.55	116.82
单位质量压缩功 / Specific compression work w / kJ·kg^{-1}	24.91	28.92
制冷性能系数 / Coefficient of performance COP	5.52	4.04

代入计算 / Substitution

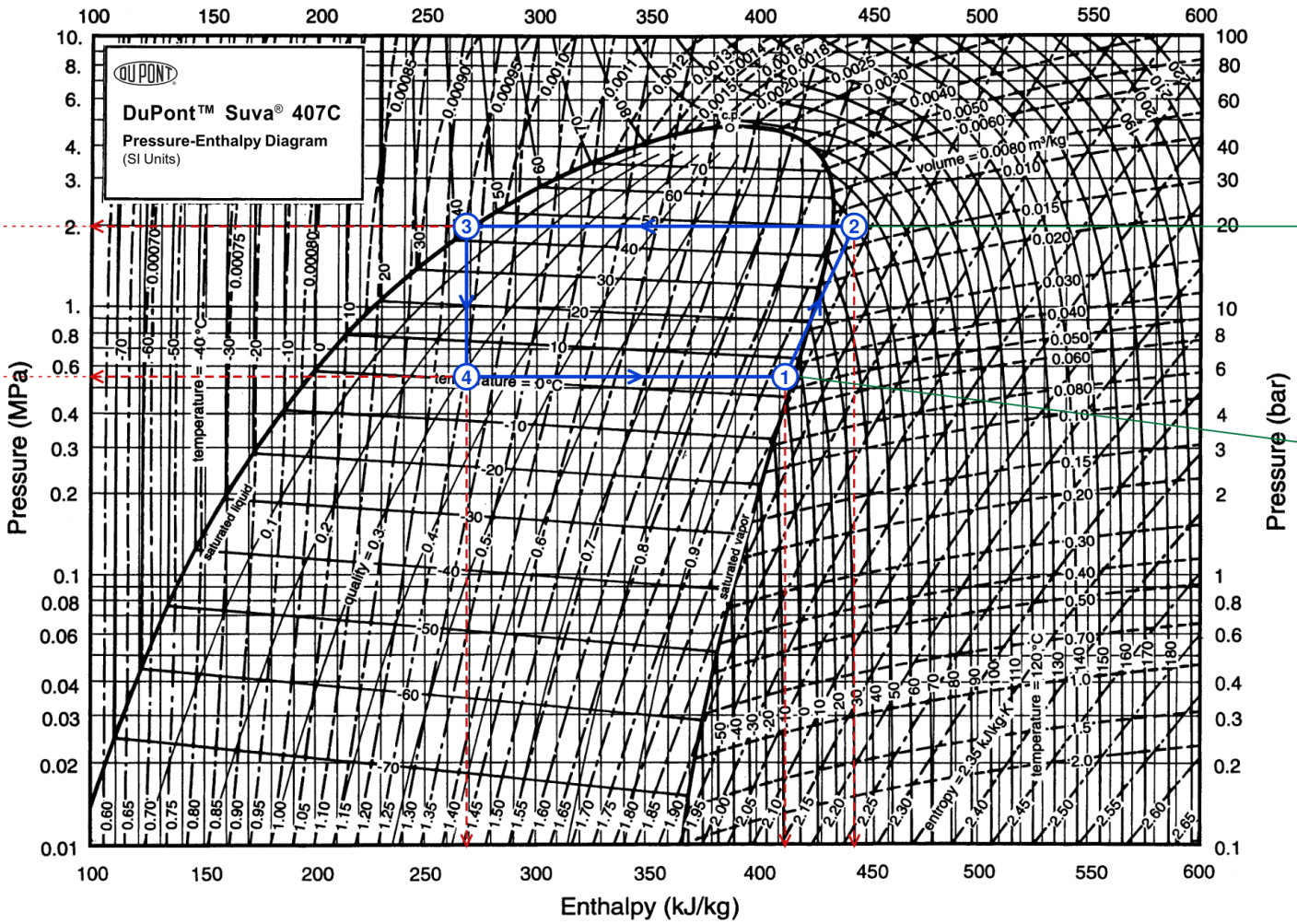
计算 / Calculation	**Chiller**	**Room cooler**
⑤ 制冷效应 / Refrigeration effect q_e = h1 − h3	401.49 - 263.94 = 137.55 kJ/kg	404.32 - 287.50 = 116.82 kJ/kg
⑥ 压缩功 / Compressor work w = h2 − h1	426.40 - 401.49 = 24.91 kJ/kg	433.24 - 404.32 = 28.92 kJ/kg
⑦ COP COP = q_e / w	137.55 / 24.91 = 5.52	116.82 / 28.92 = 4.04

图为近似读数，物性辅助核对；计算用表列焓值。 / Approximate plot readings, property checked; arithmetic uses the displayed enthalpies.

【7】R407C　冷水机组 / Chiller　Te = 5℃，Tc = 45℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.



pc = 1.9722 MPa / 19.722 bar

pe = 0.5469 MPa / 5.469 bar

h3=h4 = 268.64

h1 = 411.75

h2 = 442.77

R407C | Chiller

Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

Te uses dew point.

Tc uses bubble point.

POINT 2: COMPRESSOR OUT

v2 = 0.01241 m3/kg

s2 = 1.77001 kJ/(kg.K)

POINT 1: COMPRESSOR IN

v1 = 0.04291 m3/kg

s1 = 1.77001 kJ/(kg.K)

FROM h READINGS

qe = h1-h3 = 143.11

w = h2-h1 = 31.02

COP = qe/w = 4.61

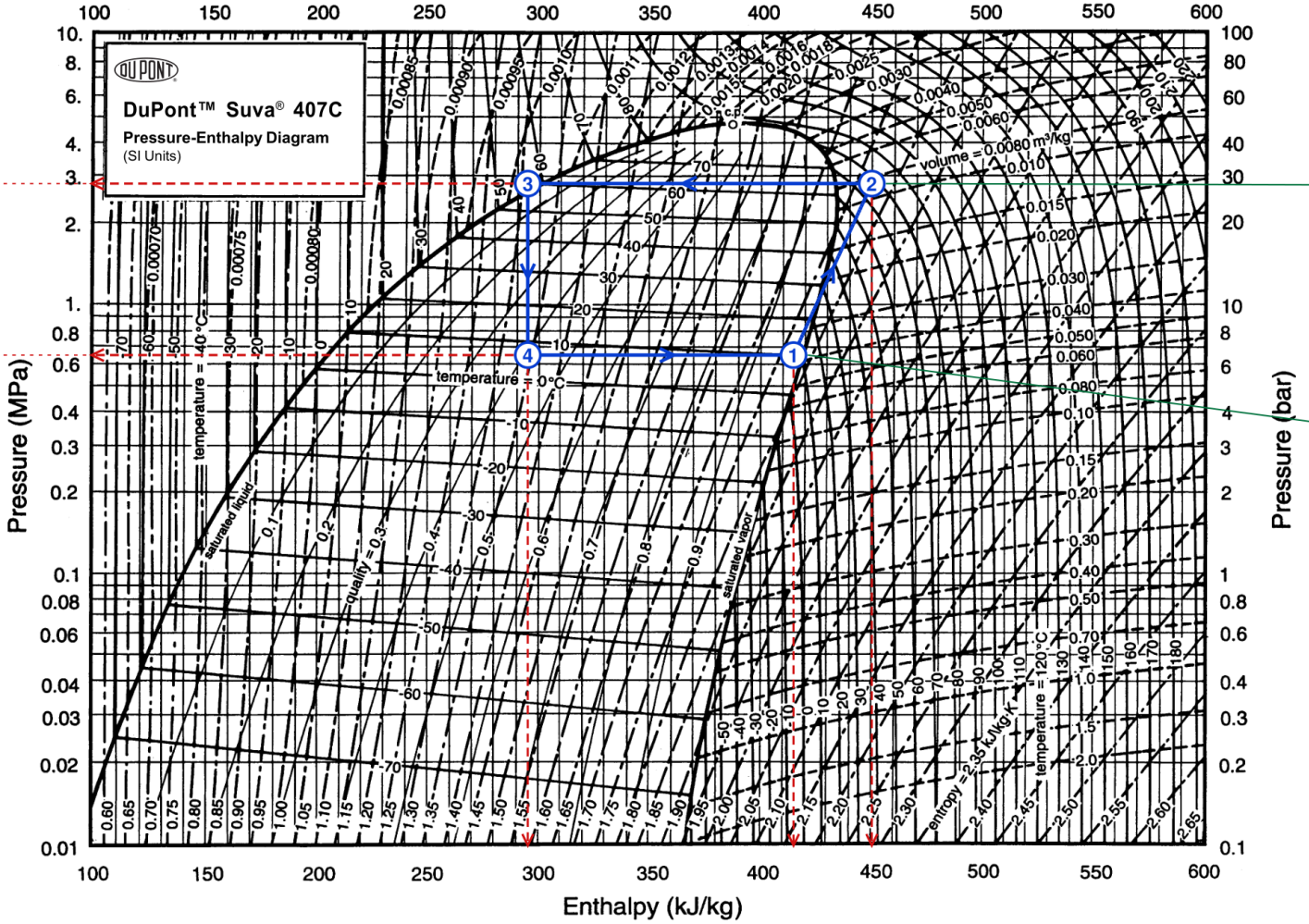
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

【8】R407C　房间冷却器 / Room cooler　Te = 10℃，Tc = 60℃

作图顺序：①定 pe、pc 并向左读压；②定点 1、3、4；③由 1 沿等熵线定点 2；④将 h1、h2、h3 竖直投影至底部焓轴（h4 = h3）。

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy.　RED DASHED: pressure / enthalpy read-off guides.



pc = 2.7694 MPa / 27.694 bar

pe = 0.6449 MPa / 6.449 bar

h3=h4 = 295.14

h1 = 414.02

h2 = 449.04

R407C | Room cooler

Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

Te uses dew point.

Tc uses bubble point.

POINT 2: COMPRESSOR OUT

v2 = 0.00864 m3/kg

s2 = 1.76430 kJ/(kg.K)

POINT 1: COMPRESSOR IN

v1 = 0.03643 m3/kg

s1 = 1.76430 kJ/(kg.K)

FROM h READINGS

qe = h1-h3 = 118.88

w = h2-h1 = 35.02

COP = qe/w = 3.39

qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

R407C　读图数据与逐步计算 / Readings and calculations

计算基准：1 为饱和蒸气，3 为饱和液体；1→2 等熵，h4 = h3；无额外过热／过冷。压力均为绝对压力。

Basis: saturated vapour at 1, saturated liquid at 3; 1→2 isentropic; h4 = h3; no extra superheat/subcooling. Pressures are absolute.

R407C：Te 采用蒸发露点，Tc 采用冷凝泡点 / Te uses evaporating dew point; Tc uses condensing bubble point.

读数及结果 / Quantity	**冷水机组 / Chiller** **5℃ / 45℃**	**房间冷却器 / Room cooler** **10℃ / 60℃**
蒸发压力 / Evaporating pressure p_e / bar abs	5.469	6.449
冷凝压力 / Condensing pressure p_c / bar abs	19.722	27.694
入口焓 / Suction enthalpy h1 / kJ·kg^{-1}	411.75	414.02
出口焓 / Discharge enthalpy h2 / kJ·kg^{-1}	442.77	449.04
液体及节流出口焓 / Liquid and throttle outlet h3 = h4 / kJ·kg^{-1}	268.64	295.14
入口比容 / Inlet specific volume v1 / m^3·kg^{-1}	0.04291	0.03643
出口比容 / Outlet specific volume v2 / m^3·kg^{-1}	0.01241	0.00864
入口、出口熵 / Inlet and outlet entropy s1 = s2 / kJ·kg^{-1}·K^{-1}	1.77001	1.76430
制冷效应 / Refrigeration effect q_e / kJ·kg^{-1}	143.11	118.88
单位质量压缩功 / Specific compression work w / kJ·kg^{-1}	31.02	35.02
制冷性能系数 / Coefficient of performance COP	4.61	3.39

代入计算 / Substitution

计算 / Calculation	**Chiller**	**Room cooler**
⑤ 制冷效应 / Refrigeration effect q_e = h1 − h3	411.75 - 268.64 = 143.11 kJ/kg	414.02 - 295.14 = 118.88 kJ/kg
⑥ 压缩功 / Compressor work w = h2 − h1	442.77 - 411.75 = 31.02 kJ/kg	449.04 - 414.02 = 35.02 kJ/kg
⑦ COP COP = q_e / w	143.11 / 31.02 = 4.61	118.88 / 35.02 = 3.39

图为近似读数，物性辅助核对；计算用表列焓值。 / Approximate plot readings, property checked; arithmetic uses the displayed enthalpies.